

DILLON, MT, USA

WMO: 727700

Lat: 45.258N

Lon: 112.554W

Elev: 5200

StdP: 12.14

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 24138

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-9.3	-2.1	-15.8	2.8	-9.0	-8.7	4.2	-1.2	28.1	31.1	25.6	32.4	6.7	50	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.1	88.4	59.4	85.3	58.6	82.2	57.6	62.4	81.7	60.8	79.8	59.3	77.8	9.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.5	79.6	68.1	53.9	75.0	66.8	51.9	69.7	65.6	30.6	81.7	29.2	79.9	28.1	77.7	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.4	23.3	20.0	DB	-17.5	93.8	7.3	2.2	-22.8	95.4	-27.0	96.7	-31.1	97.9	-36.5	99.5	(4)
(5)			WB	-17.9	67.6	7.2	3.9	-23.1	70.4	-27.3	72.7	-31.4	74.9	-36.6	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	24.4	25.7	34.6	41.5	50.2	57.9	67.2	64.8	56.2	43.9	32.1	22.1		
	DBStd	17.62	10.45	10.96	9.77	7.72	7.79	7.05	4.81	5.16	7.91	9.20	11.53	10.93		
	HDD50	3998	795	679	480	272	95	16	0	0	32	226	539	864		
	HDD65	8041	1260	1099	943	706	460	230	33	65	275	653	988	1329		
	CDD50	1620	0	0	2	16	102	251	532	460	218	38	1	0		
	CDD65	188	0	0	0	0	2	15	100	60	11	0	0	0		
	CDH74	3642	0	0	0	2	85	381	1638	1190	333	13	0	0		
	CDH80	1166	0	0	0	0	11	103	590	387	74	1	0	0		
	WSAvg	9.0	10.2	9.9	10.0	9.7	9.0	8.1	7.5	7.5	8.1	8.8	9.4	9.2		
	PrecAvg	11.5	0.4	0.2	0.6	1.3	2.1	2.0	1.1	1.2	1.0	0.7	0.4	0.3		
(15) Precipitation	PrecMax	19.2	1.0	0.9	1.8	3.1	4.9	4.9	4.3	3.3	3.3	2.2	1.4	1.2		
	PrecMin	5.7	0.0	0.0	0.0	0.1	0.2	0.3	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.0	0.3	0.2	0.4	0.7	1.1	1.1	0.9	0.7	0.8	0.6	0.3	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	52.7	64.4	72.3	81.2	88.2	92.8	91.1	86.2	75.2	62.7	47.3		
		MCWB	37.7	39.1	44.0	48.6	54.0	58.7	61.8	59.0	56.9	51.4	45.1	38.0		
	2%	DB	43.4	47.3	58.6	67.3	75.7	82.5	89.6	88.0	82.0	69.5	56.0	43.2		
		MCWB	35.2	36.9	41.7	46.4	52.3	57.7	60.6	58.4	55.4	49.1	42.6	35.6		
	5%	DB	39.9	43.3	53.9	62.1	71.7	78.6	86.2	84.3	78.4	64.1	51.6	39.4		
		MCWB	33.2	34.5	40.1	44.3	51.3	56.8	59.8	57.4	53.8	46.8	40.3	33.0		
	10%	DB	37.1	39.5	49.4	56.8	66.2	73.7	83.0	81.3	73.2	59.3	46.7	36.4		
		MCWB	31.3	32.2	37.8	41.8	49.1	54.8	58.9	56.7	52.4	45.1	37.8	31.0		
	0.4%	WB	39.2	40.2	45.6	50.2	57.2	63.2	66.3	63.3	60.4	53.4	46.9	38.9		
		MCDB	46.1	50.7	61.4	68.8	73.7	81.9	86.3	84.6	77.4	70.9	58.6	46.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	37.4	42.8	47.3	54.3	60.1	63.4	60.7	57.6	50.4	43.3	36.3		
		MCDB	42.6	45.6	56.7	64.0	71.1	78.0	82.3	80.1	76.2	66.1	54.8	42.5		
	5%	WB	33.6	35.1	40.7	45.0	52.2	57.8	61.8	59.3	55.4	48.0	40.8	33.6		
		MCDB	39.5	42.7	52.3	60.0	68.2	74.2	81.0	78.4	73.0	62.0	50.4	39.1		
	10%	WB	31.8	32.8	38.4	42.8	50.1	56.0	60.3	57.8	53.5	45.6	38.3	31.3		
		MCDB	37.0	39.1	48.5	55.5	64.8	71.6	79.2	77.0	70.3	58.3	46.5	36.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.8	18.6	21.2	23.1	24.4	26.5	32.1	31.8	28.1	23.7	19.4	16.8		
		MCDBR	18.1	21.1	26.5	31.6	32.4	34.1	36.4	36.7	35.4	30.6	23.9	17.5		
		MCWBR	12.4	13.3	14.3	15.1	14.2	14.9	15.5	15.5	15.4	15.7	13.5	12.3		
		MCDBR	17.1	19.5	24.7	29.3	29.3	31.3	32.9	32.7	31.1	28.5	22.5	17.2		
	5% WB	MCWBR	12.1	13.0	13.9	14.6	13.4	14.6	15.0	14.5	14.5	15.0	13.3	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.236	0.244	0.262	0.297	0.309	0.318	0.327	0.328	0.300	0.273	0.252	0.235		
	taud		2.411	2.402	2.493	2.450	2.492	2.500	2.497	2.461	2.539	2.587	2.547	2.470		
	Ebn at Noon		287	305	312	306	304	300	296	292	295	291	280	277		
	Edh at Noon		22	27	29	33	33	33	33	33	28	23	19	19		
(44) All-Sky Solar Radiation	RadAvg		514	826	1235	1609	1842	2138	2271	1956	1490	955	580	417		
	RadStd		49	70	77	118	172	170	117	107	123	75	45	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	(e) -5.83	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.19	N/A	N/A	N/A	N/A

Nomenclature: See separate page